

Grammatical Error Correction on C4_200M

Bag-of-Words → LSTM → Attention → Transformer

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The Problem: Grammatical Error Correction

Rewrite an ungrammatical sentence into its corrected form.

Example

Input (corrupted): *"She go to school yesterday and learning many thing."*

Output (corrected): *"She went to school yesterday and learned many things."*

- Errors: verb tense, agreement, articles, plurals, punctuation, spelling.
- The output is a **whole new sentence** — a sequence-to-sequence task.

The Data: C4_200M Subset

C4_200M — clean web text automatically corrupted into (corrupted, clean) pairs.

Split	Pairs
Train	850,000
Validation	50,000
Test	100,000
Total	1,000,000

- 16k Byte-Level BPE tokenizer.
- Max length 64 tokens.
- Same splits + tokenizer for every model (no leakage).

Four Models of Increasing Power

#	Model	What it adds
1	Bag-of-Words (TF-IDF)	Baseline — only <i>detects</i> errors
2	LSTM encoder-decoder	First corrector; fixed-size bottleneck
3	LSTM + Bahdanau attn	Decoder re-queries the full source
4	Transformer	Self-attention; no recurrence

Model 1 *detects*. Models 2–4 *correct*.

Model 1 — Bag-of-Words Baseline

TF-IDF (unigrams+bigrams, 50k vocab) + Logistic Regression / Linear SVM. Each pair → two samples: corrupted = 0, clean = 1.

Result: ~62 % accuracy — barely above 50 % chance

Why so weak? It is *structural*, not bad tuning:

- **Detector, not corrector** — outputs one label, can never produce corrected text.
- **Bag-of-words discards order** — but grammar *is* order (“she goes” vs “go she”).
- **Error signal swamped** — a 1-word error is invisible next to content-word TF-IDF weights.
- **No generalisation** — only memorises lexical statistics.

Proves GEC needs sequence modelling + generation.

Model 2 — LSTM Encoder–Decoder

- BiLSTM encoder compresses input into **one** 512-dim state.
- LSTM decoder generates the correction token by token.
- Teacher forcing $1.0 \rightarrow 0.5$; 19.55 M params; 3 epochs.

The bottleneck

After the first step the decoder never “sees” the source again — everything is squeezed into 512 numbers.

Test GLEU **0.213** (greedy) — weakest corrector.

Model 3 — LSTM + Bahdanau Attention

Keep *all* encoder states; the decoder attends to them every step.

$$\text{score}(s_t, h_i) = v^\top \tanh(W_s s_t + W_h h_i), \quad \alpha = \text{softmax}(\text{score}), \quad c_t = \sum_i \alpha_{t,i} h_i$$

- Removes the bottleneck; alignment weights are interpretable.
- Padding masked with $-\infty$ before softmax.
- 29.06 M params.

Result — the decisive jump

GLEU **0.213** → **0.599** (+0.386). Attention matters most.

Model 4 — Transformer

- No recurrence — self-attention only.
- $d_{\text{model}}=256$, 8 heads, 3+3 layers, FFN 512.
- Sinusoidal positional encoding; causal mask in decoder.
- Label smoothing 0.1; warmup LR; weight tying.

Self-attention: every token directly attends to every other — $O(1)$ path length, fully parallel.

Smallest & strongest

Only **8.05 M** params ($3.6\times$ fewer than Model 3).

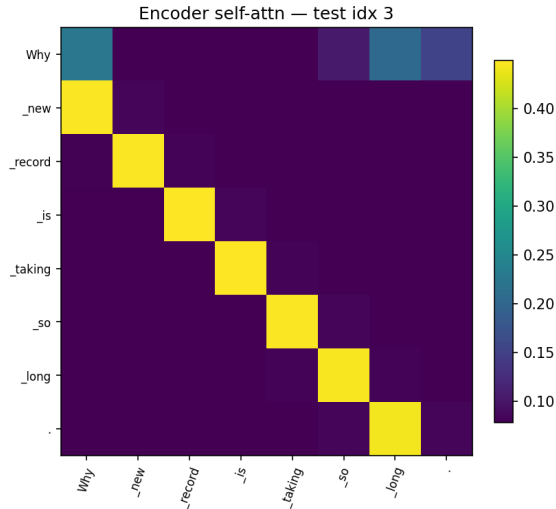
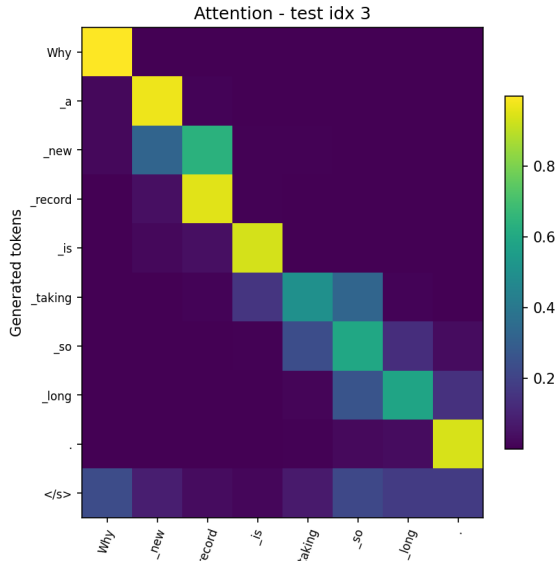
Best validation GLEU **0.613**.

Comparative Results — Test Set

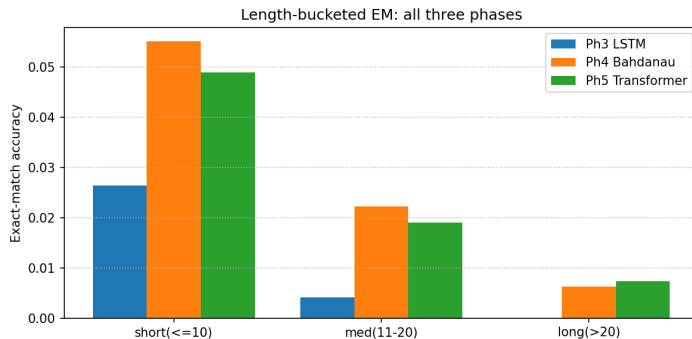
Model	Decoding	Params	EM	GLEU
LSTM (no attention)	greedy	19.55 M	≈ 0.008	0.213
LSTM (no attention)	beam-4	19.55 M	–	0.256
LSTM + Bahdanau	greedy	29.06 M	0.024	0.599
LSTM + Bahdanau	beam-4	29.06 M	0.026	0.618
Transformer	greedy	8.05 M	0.022	0.618
Transformer	beam-4	8.05 M	0.020	0.618

- Beam search helps the LSTM, not the (well-calibrated) Transformer.
- Transformer ties Model 3 on GLEU with $3.6\times$ fewer parameters.

Attention Is Interpretable



Behaviour by Sentence Length



- Bottleneck LSTM (blue) collapses to ≈ 0 on medium/long sentences.
- Bahdanau & Transformer stay non-zero everywhere — attention is what prevents the collapse.
- **Crossover:** Transformer (green) overtakes Bahdanau (orange) on the *long* bucket — its $O(1)$ path helps most there.

Worked Example — The Three Correctors

Short sentence (insert a missing article):

SRC : Why new record is taking so long.

REF : Why the new record is taking so long.

P3 : Why new record is taking so long. LSTM - no edit (copies)

P4 : Why a new record is taking so long. Bahd. - inserts article "a"

P5 : Why new record is taking so long? Trans.- "." -> "?"

Long, noisy sentence:

P3 : "...cpue to buy and/indukine stead..." hallucinated nonsense

P4 : copies source, corrupts the URL

P5 : copies source, fixes "got" -> "get" the one real fix

LSTM collapses on long input; P4 & P5 differ by 1–2 tokens.

Why Bahdanau LSTM $\approx$ Transformer

Test GLEU is tied at **0.618**. The Transformer *matches*, not beats:

- 1 **Decisive feature is shared** — both have full source access via attention (the +0.386 jump). LSTM-vs-self-attention is second-order.
- 2 **Both hit the data ceiling** — same budget, same noisy C4_200M references $\rightarrow$ both converge to GLEU ≈ 0.62 .
- 3 **Sentences are mostly short** — the Transformer's $O(1)$ path only helps on long-range deps (see the crossover).
- 4 **On hard inputs both just copy** — outputs differ by 1–2 tokens.

The real win is efficiency

Same quality with **8.05 M** params vs 29.06 M ($3.6\times$ fewer) + faster parallel training — it would pull ahead with more data / cleaner refs.

Grammar-Rule Analysis of Outputs

Learned well (local syntax):

- Subject-verb agreement: “*What are this job*” → “*What is this job*”
- Articles: “*Why new record*” → “*Why the new record*”
- Verb tense, punctuation, spacing.

Still hard:

- Long-range / semantic edits; rare-word spelling (hallucination).
- Dominant failure = *under-correction*: copying the source unchanged (the safe, high-probability action).

The Final Challenge — Dataset Quality

Why does **every** model score only $\sim 2\%$ exact match?

Reference quality is the ceiling — not model capacity

1. **References are paraphrases:** “*informed*” $\rightarrow$ “*suggest*”, “*startups*” $\rightarrow$ “*start-ups*” — not grammar errors.
2. **References can be noisy:** target “*no brain worky*” — “*worky*” is not a word.
3. **Many valid corrections exist;** EM credits exactly one.

$\rightarrow$ Report GLEU / $F_{0.5}$, not EM. Cleaner data would help more than a bigger model.

- **BoW** detects grammaticality at $\sim 62\%$ — cannot correct.
- **LSTM** corrects but is limited by the bottleneck (GLEU 0.213).
- **Bahdanau attention** is the decisive jump (GLEU 0.599).
- **Transformer** ties on GLEU (0.618) with $3.6\times$ fewer params and best validation GLEU (0.613).

Key insight

The binding constraint is **dataset quality**, not architecture — synthetic C4_200M references are often paraphrases or noise.

Thank you
Questions?